YEAR 8 MATHS · CHAPTER 4 · MIND MAP

Measurement and Pythagoras

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1 Length and perimeter

Perimeter is the distance around the outside.

1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm

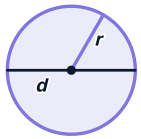
Rectangle 8 by 3: $P = 2(8 + 3) = 22$

2 Circumference

$C = \pi d = 2\pi r$

$d = 10$: $C = 10\pi \approx 31.4$

$r = 3$: $C = 6\pi \approx 18.8$



d = diameter
r = radius
d = 2r

3 Area

Rectangle lw. Triangle $\frac{1}{2}bh$.

$6 \times 4 = 24$ $\frac{1}{2} \times 8 \times 5 = 20$

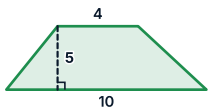
1 m² = 10 000 cm²

4 Special quadrilaterals

Parallelogram bh. Rhombus or kite $\frac{1}{2}xy$. Trapezium $\frac{1}{2}(a + b)h$.

Trapezium 4, 10, height 5: 35

Rhombus, diagonals 6 and 8: 24



5 Area of circles

$A = \pi r^2$. Use the radius, half the diameter.

$r = 5$: $A = 25\pi \approx 78.5$

$d = 8$: $r = 4$, so $A = 16\pi \approx 50.3$

Watch out: πr^2 is area, $2\pi r$ is circumference.

6 Sectors and composite shapes

Sector: $\frac{\theta}{360} \times \pi r^2$. Composite: split into shapes you know, then add or take away.

Quarter circle $r = 4$: $\frac{1}{4} \times 16\pi = 4\pi \approx 12.6$

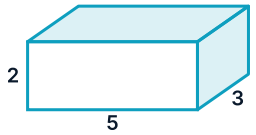
8 by 4 rectangle with a semicircle on the 4 side: $32 + \frac{1}{2}\pi(2^2) \approx 38.3$



7 Surface area of prisms

Add the areas of every face. Drawing the net helps.

Box 5 by 3 by 2: $2(15 + 10 + 6) = 62$



8 Volume and capacity

Box: $V = lwh$.

$5 \times 3 \times 2 = 30$ cm³

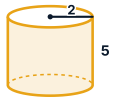
1 cm³ = 1 mL, 1000 cm³ = 1 L, 1 m³ = 1000 L

9 Volume of prisms and cylinders

$V = Ah$. Cylinder: $V = \pi r^2 h$.

Triangular prism: $\frac{1}{2} \times 3 \times 4 = 6$, length 10: $V = 60$

Cylinder $r = 2,\ h = 5$: $V = 20\pi \approx 62.8$



10 Time and time zones

Going west, clocks are earlier. Going east, clocks are later.

Melbourne is UTC+10, Perth is UTC+8

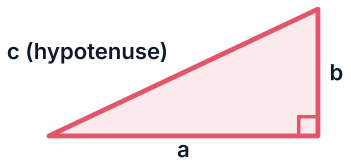
3 pm in Melbourne is **1 pm** in Perth (standard time)

Watch out: daylight saving changes the gap in summer.

11 Pythagoras' theorem

Right-angled triangles: $c^2 = a^2 + b^2$, c is the hypotenuse.

Sides 5 and 12: $c^2 = 25 + 144 = 169$, so $c = 13$



12 Finding a shorter side

Subtract: $a^2 = c^2 - b^2$.

Hypotenuse 10, side 6: $a^2 = 100 - 36 = 64$, so $a = 8$

Watch out: a shorter side can never be longer than the hypotenuse.